

Variables and Missing-Number Equations Mission

Warm-up and worked example

Name: _____

Date: _____

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YOUR MISSION Use a letter or box to represent an unknown and solve equations by reasoning.

SMART MOVE Replace the unknown and check that both sides are equal.

Math Talk

What number makes $n + 37 = 92$ true?

My idea:

New Words

expression	equation	variable
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

1. Discuss the equal sign as "has the same value as."
2. Solve $n + 7 = 19$.

What the example shows: _____

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Learn it and show your thinking

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YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** Discuss the equal sign as "has the same value as."

[] **Step 2:** Solve $n + 7 = 19$.

[] **Step 3:** Solve $4 \times n = 36$.

[] **Step 4:** Solve $48 \div n = 6$.

[] **Step 5:** Check each value by substitution.

My model, drawing, or organized work:

The big idea in my own words:

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Practice lab

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YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. $n - 28 = 47$

2. $6 \times n = 54$

3. $72 \div n = 8$

4. $n \div 5 = 7$

5. Write an equation for a word story.

Choose one answer and prove it another way:

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Challenge and final check

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YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

Find all whole-number pairs that satisfy $a \times b = 48$.

My plan:

My solution and proof:

Final Check

Explain why the letter does not always mean the same number in different equations.

☐ I can teach it

☐ I need practice

☐ I need help

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Reusable math tool

Name: _____

Date: _____

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YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Replace the unknown and check that both sides are equal.

Grid Lab

What does your drawing prove?
